Emotion Analysis Using BRIGHTER + EthioEmo Datasets
Evaluating Data Augmentation Techniques for Multilabel Emotion Classification in Low-Resource African Languages

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1 Introduction

Emotion analysis (EA) automatically identifies textual emotions, but African languages remain underrepresented due to linguistic diversity and data scarcity. Consequently, models struggle with multilabel classification, where a single text conveys multiple overlapping emotions. To mitigate this, this project investigates data augmentation, specifically back-translation and paraphrasing, to artificially expand training datasets. By evaluating these techniques alongside qualitative error analysis, this research seeks to improve multilabel emotion classification for low-resource African languages and build more inclusive, responsible NLP systems.

2 Background

Emotion analysis (EA) identifies specific, often overlapping emotional states (e.g., anger, joy) rather than basic positive or negative feelings, requiring multilabel classification for real-world accuracy (del Arco et al., 2024). However, the vast annotated data required by modern EA models creates a barrier for African languages, which remain heavily low-resource (Adebara and Abdul-Mageed, 2022). While recent Afrocentric efforts have advanced foundational NLP, complex tasks like emotion classification remain under-explored and rarely optimised for typological nuances like complex morphology and code-switching (Adebara, 2024).

To address data scarcity, researchers use data augmentation, specifically back-translation and paraphrasing, to artificially expand training sets. These approaches excel in low-resource environments by increasing lexical diversity and maintaining semantic fidelity, often outperforming AI models without prior examples (Radlinski et al., 2025). However, their application to multilabel emotion classification in African languages remains unexplored. Despite benchmarks like the EthioEmo dataset, the field is still vastly under-resourced. This project bridges this gap by evaluating back-translation and paraphrasing on multilabel emotion tasks for underrepresented African languages, providing essential insights to build inclusive NLP models.

3 Research Questions

1. How can data augmentation strategies (such as back-translation and paraphrasing) improve multilabel emotion classification for African languages?
2. How can existing NLP methods be adapted or applied effectively in low-resource African language contexts for emotion analysis?

4 Dataset and Data Description

4.1 Primary Dataset: BRIGHTER

This project uses the BRIGHTER dataset, a multilingual collection of approximately 100,000 emotion-annotated text instances across 28 languages. Each instance is labelled with six emotions (joy, sadness, anger, fear, surprise, disgust) and a neutral class, using a four-level intensity scale. The dataset supports multilabel classification, which is the focus of this study.

4.2 Selected Languages

Three African languages are selected: Afrikaans, Hausa, and Igbo. These were chosen based on data availability, societal relevance, and team familiarity. All have complete train, development, and test splits, enabling augmentation experiments. The languages represent distinct language families and together cover a large speaker base, supporting broader applicability of results.

4.3 Confirmed Availability

The dataset is publicly available on Hugging Face under a CC-BY 4.0 licence, and no additional data collection is required.

5 Approach and Methodology

5.1 Overview

This project investigates whether data augmentation improves multilabel emotion classification for low-resource African languages. The methodology consists of baseline model training, data augmentation, augmented model training, and qualitative error analysis.

5.2 Baseline Models

Two pretrained multilingual models are fine-tuned: XLM-RoBERTa-large (general multilingual baseline) and AfroXLMR-large (adapted for African languages). Models are trained separately per language using sigmoid outputs and binary cross-entropy loss for multilabel classification.

5.3 Data Augmentation Techniques

Two augmentation strategies are applied:

1. **Back-translation:** translating text to a pivot language and back to introduce lexical diversity.

2. **Paraphrasing:** generating structurally varied versions of text while preserving meaning.

Training data is expanded to approximately 1.5× and 2× its original size.

5.4 Experimental Conditions

Four conditions are evaluated:

1. Baseline (no augmentation)

2. Back-translation

3. Paraphrasing

4. Combined back-translation and paraphrasing

5.5 Error Analysis

Qualitative error analysis is conducted on the best-performing models. Misclassifications are examined to identify patterns such as emotion overlap, linguistic complexity, and class imbalance, with interpretation informed by language-specific knowledge.

6 Experiments and Results

6.1 Experimental Setup

All experiments are conducted across three language subsets of the BRIGHTER dataset: **Afrikaans**, **Hausa**, and **Igbo**. For Afrikaans, the dataset contains 1,222 training instances, 196 development instances, and 2,130 test instances (Muhammad et al., 2025a,b). For Hausa, the splits consist of 2,145 original training instances (expanded to 2,945 rows post-deduplication handling), 712 development instances, and 2,160 test instances (Muhammad et al., 2025a,b). For Igbo, the original splits provide 2,880 training instances, 958 development instances, and 2,888 test instances; 800 instances were transferred from the test split to training to mitigate the small training set size, yielding a final training pool of 3,680 instances (Muhammad et al., 2025a,b). Each instance is annotated with up to six binary emotion labels: anger, disgust, fear, joy, sadness, and surprise. For Igbo, the `surprise` label was excluded from the label space due to near-zero prevalence (2.6% of training instances), and a `neutral` class was derived implicitly for instances carrying no emotion label.

The training data profiles across these languages exhibit pronounced class imbalance. In the Hausa training partition, for example, sadness is the most dominant emotion (mean label rate ≈ 0.30), followed by anger (≈ 0.19), surprise (≈ 0.16), disgust (≈ 0.15), fear (≈ 0.15), and joy (≈ 0.15). The Afrikaans subset presents joy as the most frequent emotion (mean label rate ≈ 0.43), while surprise labels are entirely absent from its training split. For Igbo, anger is the most prevalent class (≈ 0.20), followed by disgust (≈ 0.19), sadness (≈ 0.17), joy (≈ 0.16), fear (≈ 0.08), and surprise (≈ 0.03) (Muhammad et al., 2025a,b). Neutral instances, derived from all-zero label rows, account for approximately 21.5% of the Igbo training set across all splits.

Two pretrained multilingual models are evaluated: **XLM-RoBERTa-large** and **AfroXLMR-large**. Both models are fine-tuned for five epochs using sigmoid outputs with binary cross-entropy loss for multilabel classification. Performance is reported using macro, micro, and weighted F1 scores on the held-out test sets, with macro F1 serving as the primary evaluation metric owing to its sensitivity to underrepresented and minority emotion classes.

Four experimental conditions are systematically evaluated per language:

1. **Condition A – Baseline:** No augmentation; models are trained exclusively on the original human-annotated datasets (1,222 instances for Afrikaans; 2,945 instances for Hausa; 3,680 instances for Igbo).

2. **Condition B – Back-translation:** Training data is expanded by adding high-resource pivot translations. For Afrikaans, this totals 3,030 instances via Dutch and English pivots. For Hausa, the training set is expanded to 4,417 instances by adding English (736 rows) and French (736 rows) back-translations. For Igbo, English and French pivots via NLLB-200 produce 897 and 910 usable augmented rows respectively, expanding the training set to 5,487 instances.

3. **Condition C – Paraphrase-only:** Training sets are expanded via stochastic paraphrasing. Afrikaans is expanded to 3,033 instances. For Hausa, the corpus is expanded to 4,417 instances by adding 1,472 stochastically paraphrased rows. For Igbo, paraphrasing via an Ollama-served LLM generates 1,836 additional rows, expanding the training set to 5,516 instances.

4. **Condition D – Combined:** The maximal data condition combining all augmentation paths. This expands Afrikaans to 4,041 instances, Hausa to 5,889 total training instances, and Igbo to 7,346 total training instances (Original + English BT + French BT + Paraphrasing).

6.2 Results and Cross-Lingual Evaluation

Tables 1, 2, and 3 summarise test set performance across all conditions for Afrikaans, Hausa, and Igbo respectively. For Afrikaans, two runs produced anomalous near-zero scores due to suspected runtime label-alignment faults, specifically Condition B and C for AfroXLMR-large, and are omitted from analysis. For Hausa, all pipelines executed successfully across all conditions. For Igbo, all conditions completed successfully; however, the initial Condition D AfroXLMR-large run collapsed to near-zero F1 due to training instability under the heterogeneous 4x augmented corpus and required a full retraining run, which stabilised at a macro F1 of 0.4554 by epoch 5.

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.5996	0.6396	0.6492
A – Base.	AfroXLMR	0.6333	0.6550	0.6603
B – BT	XLM-R	0.5732	0.5977	0.6019
C – Para.	XLM-R	0.6768	0.7074	0.7134
D – Comb.	XLM-R	0.6531	0.6802	0.6862
D – Comb.	AfroXLMR	**0.7302**	**0.7632**	**0.7661**

Table 1: Test F1 scores (Afrikaans). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large. BT = back-translation; Para. = paraphrase. Conditions B & C (AfroXLMR) omitted due to a suspected runtime error.

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.5944	0.6109	0.6100
A – Base.	AfroXLMR	0.6793	0.6902	0.6896
B – BT	XLM-R	0.6403	0.6488	0.6492
B – BT	AfroXLMR	0.6003	0.6176	0.6140
C – Para.	XLM-R	0.6318	0.6418	0.6425
C – Para.	AfroXLMR	**0.7125**	**0.7206**	**0.7198**
D – Comb.	XLM-R	0.6458	0.6567	0.6545
D – Comb.	AfroXLMR	0.7010	0.7068	0.7077

Table 2: Test F1 scores (Hausa). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large. Epoch validation limits are reported as proxy parameters where final test checkpoints are optimised.

Baseline performance. Across all languages evaluated, both architectures establish highly competitive baselines. AfroXLMR-large consistently outclasses XLM-RoBERTa-large under baseline conditions (Condition A). In Afrikaans, AfroXLMR achieves a macro F1 of 0.6333 versus 0.5996 for XLM-RoBERTa-large. In Hausa, it reaches 0.6793 versus 0.5944. In Igbo, AfroXLMR-large achieves 0.4858 versus 0.4500 for XLM-RoBERTa-large. The lower absolute scores for Igbo reflect the greater linguistic distance from high-resource languages in the pretraining corpora, as well as the prevalence of code-switching and informal social media text in the Igbo subset. This recurrent trend across all three languages confirms that AfroXLMR's language-adapted pre-training provides a superior, linguistically grounded starting point for African morphosyntax and vocabulary representations.

Effect of back-translation. The downstream impact of back-translation reveals substantial architectural disparities. For XLM-RoBERTa-large, back-translation significantly lifts Hausa performance, jumping from a baseline macro F1 of 0.5944 to 0.6403 (+4.59 points). Conversely, it drops Afrikaans performance by 2.64 points (0.5996 $\rightarrow$ 0.5732). For AfroXLMR-large, back-translation causes severe performance degradation in Hausa, plunging its macro F1 score to 0.6003. In

Cond.	Model	Mac	Mic	Wt
A – Base.	XLM-R	0.4500	0.5361	0.5168
A – Base.	AfroXLMR	0.4858	0.5663	0.5563
B – BT	XLM-R	0.4347	0.5204	0.4941
B – BT	AfroXLMR	0.4855	0.5621	0.5570
C – Para.	XLM-R	0.4841	0.5396	0.5404
C – Para.	**AfroXLMR**	**0.5045**	**0.5841**	**0.5787**
D – Comb.	XLM-R	0.4925	0.5584	0.5572
D – Comb.	AfroXLMR	0.4554	0.5288	0.5245

Table 3: Test F1 scores (Igbo). Mac/Mic/Wt = macro/micro/weighted F1. XLM-R = XLM-RoBERTa-large; AfroXLMR = AfroXLMR-large. BT = back-translation; Para. = paraphrase. Condition D (AfroXLMR) reflects a retrained run after an initial unstable convergence.

Igbo, back-translation yields mixed results: XLM-RoBERTa-large declines modestly from 0.4500 to 0.4347, while AfroXLMR-large is largely unaffected (0.4858 to 0.4855), suggesting that the Africa-adapted model is more robust to the noise introduced by English and French pivot translations. This suggests that routing text through intermediate high-resource pivot configurations can introduce detrimental semantic noise and label drift, particularly for general-purpose multilingual models.

Effect of paraphrasing. Paraphrasing proves to be a universally powerful and highly stable augmentation strategy. For XLM-RoBERTa-large in Afrikaans, it produces a +7.72 point improvement over baseline (0.5996 $\rightarrow$ 0.6768). For Hausa, the paraphrase-only condition (Condition C) combined with AfroXLMR-large unlocks the single highest primary metric score recorded for that language, peaking at a macro F1 of **0.7125**, an increase of 3.32 points over its baseline. For Igbo, Condition C with AfroXLMR-large achieves the best result across all Igbo conditions at a macro F1 of **0.5045**, a gain of $+1.87$ points over the AfroXLMR baseline, and XLM-RoBERTa-large also improves substantially from 0.4500 to 0.4841. This consistent pattern across all three languages indicates that generating localised surface-level variety while keeping semantic structures tightly controlled provides a cleaner, highly dependable training signal.

Effect of combined augmentation. The combined multi-source augmentation condition (Condition D) reveals cross-lingual differences. For Afrikaans, combining strategies yields the absolute best overall configuration when paired with AfroXLMR-large, achieving a peak macro F1 of 0.7302 (+9.69 points over baseline). For Hausa, stacking all augmentations together expands the training set to 5,889 samples but slightly dilutes peak performance: AfroXLMR-large reaches a macro F1 of 0.7010 under Condition D, which, while a major lift over the baseline, falls short of the isolated paraphrase-only run. In Igbo, the pattern is more pronounced: XLM-RoBERTa-large recovers to 0.4925 under Condition D, its second-best result, but AfroXLMR-large drops to 0.4554, well below its baseline. Inspection of training curves revealed that AfroXLMR-large converged to near-zero F1 in an initial unstable run and required retraining; after retraining it stabilised but was unable to match its paraphrase-only performance. Across all three languages, this behaviour demonstrates that combining heterogeneous augmentation sources can introduce optimisation instability, particularly for language-adapted models already sensitive to data distribution shifts.

6.3 Qualitative Error Analysis and Linguistic Extensions

Post-hoc qualitative error analysis was conducted on the highest-performing configurations across all three language settings (AfroXLMR-large, Condition C for Hausa and Igbo; Condition D for Afrikaans). Several prominent pattern domains emerged, tightly mapping to the original linguistic dimensions defined in Section 5.6.

Neutral class confusion and valence boundary blurring. Implicitly derived class spaces produced elevated error signatures cross-lingually, aligned with the Emotion Overlap and Ambiguity dimension. For instance, the Igbo `neutral` class generated the highest cumulative error boundaries with 388 false positives and 400 false negatives, finishing at an F1-score of 0.36 despite its high density. Because `neutral` functions as an implicit fallback rather than an explicit token, models frequently struggle to separate muted or underexpressed emotions from factual sentences. Similarly, in Afrikaans, where the emotion *surprise* was completely unrepresented in the training vector, and in Hausa, where valence thresholds are compressed, the classifier routinely defaulted to `neutral` predictions under local token ambiguity.

Linguistic complexity, code-switching, and token placeholders. Typological and orthographic adjustments altered error rates across the evaluated cohorts. Igbo social media texts frequently mix target matrices with English and Nigerian Pidgin; surprisingly, code-switched rows showed a lower error frequency (44.9%) than pure Igbo strings

(50.0%). This implies that English anchors present explicit cross-lingual tokens that the shared base embeddings can map with high reliability. Conversely, structural noise like platform URL placeholders (`##url##`) significantly degraded performance, raising the error rate to 55.7% in Igbo and causing similar text-length attenuation in Hausa, where short mean text constraints (e.g., 23.5 tokens) left insufficient contextual signal for structural resolution.

Class imbalance and multi-label collapse. Severe instance skew and sparse multilabel distributions remained a fundamental challenge across all language paths, reinforcing the Class Imbalance Distortions. Instances carrying two or more concurrent emotion labels in Igbo experienced an extreme error rate of 96.6%, compared to 47.7% for isolated single-label lines. Because combinations are highly constrained, the model lacks the data depth to learn co-occurrence maps. This led to high-frequency collapsing behaviors, such as hidden *joy* being misclassified as `neutral` in 63.8% of missed instances in Igbo, and minority emotions like *anger* or *disgust* in Afrikaans being overwhelmed by dominant classes despite the introduction of paraphrase expansions.

7 Reflections

This multi-language comparative project reveals critical insights into applying synthetic data generation to multilabel emotion classification across distinct African language families, highlighting low-resource NLP engineering and ethical constraints.

7.1 Experimental Retrospective: Successes, Failures, and Challenges

What Worked: Localised Diversity via Paraphrasing. Stochastic paraphrasing (Condition C) emerged as the most robust augmentation technique across all three languages. It delivered a +7.72 macro F1 lift for XLM-RoBERTa-large in Afrikaans, and unlocked peak configurations for Hausa (0.7125) and Igbo (0.5045) using AfroXLMR-large. By preserving semantic boundaries while generating synonyms, paraphrasing regularised the models and mitigated small-sample variance.

What Didn't Work: Multi-Pivot Back-Translation and Data Pooling. Multi-pivot back-translation (Condition B) underperformed, degrading AfroXLMR-large in Hausa and XLM-RoBERTa-large in Afrikaans. Routing text through intermediate high-resource pivots (Dutch, French, English) introduced semantic drift, flattening nuanced emotional expressions. Stacking all sources (Condition D) caused data distribution shifts, diluting Hausa results and triggering training collapse in Afrikaans and Igbo. This proves data scaling is non-linear and noise-sensitive.

Engineering Challenges: Operational Instability. The primary bottleneck was maintaining label and sequence integrity. Because multilabel targets require exact alignment between text strings and a binary flag vector, formatting anomalies or parsing offsets from the NLLB-200 decoding engine decoupled tokens from their classes. This caused runtime failures for AfroXLMR-large in Afrikaans (Conditions B and C) and complete convergence collapse in Igbo (Condition D), requiring rigorous debugging and retraining.

7.2 Responsible AI Principles and Societal Impacts

Data Openness, Access, and Reproducibility. This project strictly utilises the public BRIGHTER benchmark dataset under a CC-BY 4.0 open licence. To ensure full reproducibility without relying on proprietary APIs, our complete code repositories—including data filters, random seeds, and tokenisation dictionaries—are structured for public access, lowering technical barriers for regional researchers auditing low-resource pipelines.

Fairness and Linguistic Equity. Baseline evaluations (Condition A) exposed stark resource inequalities: AfroXLMR-large achieved macro F1 scores of 0.6333 for Afrikaans and 0.6793 for Hausa, but dropped to 0.4858 for Igbo. This reflects the systemic under-representation of Igbo in web-scale pre-training corpora. By establishing localised augmentation loops without human collection budgets, this framework advances linguistic equity without relying on massive English-centric transfer paradigms.

Limitations and Risks of Synthetic Data. Sequence-to-sequence generators carry historical biases from pre-training; routing African text through NLLB-200 risks embedding homogenised, Westernised emotional idioms. Furthermore, augmentation cannot resolve structural ambiguity. Our

Igbo multi-label instances suffered a 96.6% error rate, and the derived `neutral` class produced over 788 false assignments. Augmentation behaves as an embedding regulariser, not a replacement for expert human annotation.

Broader Societal Impacts and Inclusivity. Low-resource communities face severe model bias when content moderation or mental health screening tools misclassify local colloquialisms or code-switching. Interestingly, Igbo text mixing English and Nigerian Pidgin yielded a lower error rate (44.9%) than pure Igbo (50.0%), showing model dependency on English anchors. Natively understanding regional structures helps bridge the digital divide and ensures equitable algorithmic safety.

7.3 Future Extensions and Improvements

To resolve the identified architectural and distribution limitations, we suggest four improvements:

1. **Emotion-Stratified Oversampling:** Target synthetic generation specifically toward minority labels (e.g., *anger*, *disgust*) rather than applying uniform augmentation across all training samples.

2. **Cross-Lingual Few-Shot Prompting:** Augment data matrices with few-shot context examples drawn from typologically related, higher-resource regional languages.

3. **Automated Quality Filtering:** Integrate cross-lingual semantic similarity filters (e.g., LaBSE) to automatically discard augmented outputs exhibiting high translation drift or label mismatch before fine-tuning.

4. **Cross-Class Co-occurrence Modelling:** Replace independent binary cross-entropy projections with a loss function or label-correlation architecture that explicitly models co-occurring emotion states.

8 Conclusion

This project investigated whether data augmentation techniques–specifically multi-pivot back-translation and stochastic paraphrasing–can improve multilabel emotion classification performance for the low-resource African languages Afrikaans, Hausa, and Igbo using the BRIGHTER benchmark framework. Two pre-trained multi-lingual models, XLM-RoBERTa-large and AfroXLMR-large, were fine-tuned across four distinct experimental data boundaries.

Our empirical findings confirm that data augmentation provides meaningful performance improvements, but its ultimate success depends on matching the specific augmentation type with the right model architecture and target language. Paraphrasing proved to be the most consistent and reliable technique overall. It yielded a 7.72 point improvement for XLM-RoBERTa-large in Afrikaans, achieved the single highest test performance for Hausa when combined with AfroXLMR-large (macro F1 of **0.7125**), and produced the best Igbo result of **0.5045** under the same configuration. Back-translation showed high variance across all three languages, actively harming XLM-RoBERTa-large in Afrikaans and Igbo, and AfroXLMR-large in Hausa, due to semantic translation drift. The maximum data condition (Condition D) achieved the overall best result for Afrikaans (0.7302 macro F1), while in both Hausa and Igbo it underperformed the paraphrase-only condition, demonstrating that scaling augmentation volume beyond a critical threshold can introduce noise that outweighs the benefits of additional data diversity.

These comparative discoveries provide actionable insights for developing robust, responsible NLP systems tailored to underrepresented African languages. They show that simple, targeted augmentation pipelines can successfully mitigate data scarcity, while underscoring that severe class imbalances and multi-label label integrity require careful handling. Future work should focus on designing emotion-stratified generation pipelines for minority labels, exploring cross-lingual transfer from related language families, and establishing automated validation frameworks to preserve multi-label text integrity during generation loops.

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